WORKED SOLUTION

STEP III 2007, Question 13

Probability and Statistics • Random Walk Expectation

STEP III 2007, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

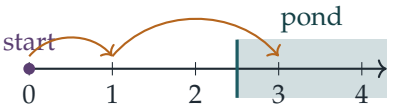
step.maths.org/questions/07-s3-q13

The key idea

The half-metre is there to remove any doubt about "landing on the edge": starting $(n - \frac{1}{2})$ m away, the frog is in the pond exactly when the total distance it has jumped is *at least* n. So everything is about the first time a running total of 1s and 2s reaches n. For small n we can simply list the possibilities; for the general formula, three values of u_n give three equations for the three constants.

Solution

Let S_j be the total distance (in metres) covered in the first j jumps, and let T be the number of jumps needed to reach the pond. Starting $(n - \frac{1}{2})$ m away, the frog is in the pond after j jumps if and only if $S_j > n - \frac{1}{2}$, that is $S_j \geq n$, since S_j is a whole number. So $p_n(j) = \mathrm{P}(T = j)$, where T is the first j with $S_j \geq n$.



The picture shows $n = 3$ (the edge is 2.5 m away) and the sequence "short, long", which reaches the pond in two jumps.

Part (i)

With $n = 2$ the edge is 1.5 m away. A long first jump (probability q) takes the frog straight in, so $p_2(1) = q$. Otherwise the first jump is short (probability p), the frog is 0.5 m from the edge, and *any* second jump takes it in. Jumps are independent, so

$$p_2(2) = p \times 1 = p,$$

as required. (Also $p_2(j) = 0$ for $j \geq 3$, and $p + q = 1$ confirms the distribution is complete.)

Part (ii)

From $\frac{1}{2}$ m away every jump lands in the pond, so $u_1 = 1$.

Using part (i),

$$u_2 = 1 \cdot p_2(1) + 2 \cdot p_2(2) = q + 2p = 2 - q.$$

For $n = 3$ we need $S_j \geq 3$.

- One jump is never enough, since $S_1 \leq 2$: $p_3(1) = 0$.
- Two jumps are not enough only if both are short ($S_2 = 2$); every other pair gives $S_2 \geq 3$. So $p_3(2) = 1 - p^2$.
- After two short jumps the frog is 0.5 m away and the third jump certainly lands in the pond: $p_3(3) = p^2$.

Therefore

$$\begin{aligned} u_3 &= 2(1 - p^2) + 3p^2 = 2 + p^2 \\ &= 2 + (1 - q)^2 = 3 - 2q + q^2, \end{aligned}$$

as required.

Ask yourself Is there a quicker way to see these values, and does it extend to all n?

Yes: condition on the first jump. After it the frog is in the same situation with n reduced by 1 or by 2, so for $n \geq 2$

$$u_n = 1 + p\,u_{n-1} + q\,u_{n-2}, \qquad u_0 = 0,$$

where $u_0 = 0$ covers a long first jump from 1.5 m. This gives $u_2 = 1 + p$ and $u_3 = 1 + p(1 + p) + q = 2 + p^2$ again. It is also why the form in part (iii) is right: the auxiliary equation $\lambda^2 - p\lambda - q = 0$ factorises as $(\lambda - 1)(\lambda + q) = 0$, giving the $(-q)^{n-1}$ term. The repeated root 1 (from the "+1") explains the $B + Cn$.

Part (iii)

Substituting $n = 1, 2, 3$ into $u_n = A(-q)^{n-1} + B + Cn$ and using part (ii):

$$A + B + C = 1, \tag{1}$$

$$-qA + B + 2C = 2 - q, \tag{2}$$

$$q^2A + B + 3C = 3 - 2q + q^2. \tag{3}$$

Subtracting to eliminate B:

$$(2) - (1): \quad -(1 + q)A + C = 1 - q, \tag{4}$$

$$(3) - (2): \quad q(1 + q)A + C = 1 - q + q^2. \tag{5}$$

Now $q \times (4) + (5)$ eliminates A:

$$(1+q)C = q - q^2 + 1 - q + q^2 = 1, \qquad \text{so} \qquad C = \frac{1}{1+q},$$

as required. Then (4) gives

$$(1+q)A = \frac{1}{1+q} - (1-q) = \frac{1-(1-q^2)}{1+q} = \frac{q^2}{1+q}, \qquad \text{so} \qquad A = \frac{q^2}{(1+q)^2},$$

and (1) gives

$$B = 1 - \frac{q^2}{(1+q)^2} - \frac{1}{1+q} = \frac{(1+q)^2 - q^2 - (1+q)}{(1+q)^2} = \frac{q}{(1+q)^2}.$$

So

$$u_n = \frac{q^2(-q)^{n-1} + q + (1+q)n}{(1+q)^2}.$$

Large n. Since $0 \le q \le 1$, the terms $A(-q)^{n-1}$ and B are bounded (each is at most $\frac{1}{4}$ in size), while Cn grows without limit. Hence for large n

$$u_n \approx Cn = \frac{n}{1+q} = \frac{n}{p+2q},$$

using $1 = p + q$, so $1 + q = p + 2q$.

Why this is expected. The mean length of one jump is $1 \cdot p + 2 \cdot q = p + 2q$ metres. The frog has to cover about n metres, and by the law of large numbers the total length of N jumps is close to $N(p+2q)$ when N is large. So the number of jumps needed is close to $n/(p+2q)$. The approximation is a good one because the half-metre start offset and the overshoot at the end (the frog lands at most 1.5 m past the edge) are bounded. They change the count by a bounded amount, which is negligible compared with n. That bounded correction is exactly the $A(-q)^{n-1} + B$ part.

Checks

- $n = 0$ in the formula gives $\left(q^2 \cdot (-1/q) + q\right)/(1+q)^2 = 0 = u_0$, as it should, and the formula satisfies the recurrence above.
- $q = 0$ (all jumps short): $u_n = n$, correct since the frog needs exactly n unit jumps.
- Exact enumeration of all jump sequences (with fractions) matched the formula for $n = 1, \ldots, 12$ and several values of p. A simulation with $q = 0.7$, $n = 200$ gave a mean of 117.90 jumps, against 117.89 from the formula and $n/(p+2q) \approx 117.65$.

Common slips

- Treating "lands in the pond" as $S_j = n$ exactly. Because of the half-metre the frog can overshoot, and every sequence with $S_j \geq n$ counts.
- Writing $p_3(2) = 2pq + q^2$ and forgetting the case where the long jump comes first. In fact $pq + qp + q^2 = 1 - p^2$ covers both orders.

Beyond STEP

The recurrence for u_n is a renewal equation. The result $u_n \sim n/\mu$, where μ is the mean step, is the elementary renewal theorem. An exact version follows from Wald's identity $\mathrm{E}(S_T) = \mu\,\mathrm{E}(T)$: here S_T is either n or $n+1$, so $(1+q)u_n = n + \mathrm{P}(\text{overshoot})$. Grimmett and Welsh, *Probability: An Introduction*, covers random walks and generating functions in this spirit.

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